Topology-Aware Federated Learning for Cyber-Physical Intrusion Detection in DER

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***Abstract*—The integration of Distributed Energy Resources (DERs) transforms traditional distribution grids into dynamic cyber-physical systems (CPS). However, the constant fluctuations in power generation caused by DERs make it easier for stealthy temporal cyberattacks, such as Replay and Denial of Service (DoS), to remain hidden within normal grid behavior. While Federated Learning (FL) provides privacy-preserving intrusion detection, existing models are fundamentally topology-blind; they evaluate edge nodes in isolation and fail to enforce underlying electrical constraints. To address this vulnerability, this paper proposes a Topology-Aware Federated Edge Learning framework. To simulate a distribution network we utilized IEEE 33-Bus network augmented with active solar DERs, the grid is geographically partitioned into distinct zones. Localized LSTM Autoencoders are deployed to edge clients to learn the temporal dynamics of physical states (Active Power, Reactive Power, Voltage Magnitude, and Phase Angle). A central server aggregates these models to enforce spatial-temporal consistency across the entire topology without centralizing raw operational data. By cross-validating physical constraints between neighboring zones, the proposed architecture successfully detects stealthy temporal manipulations that easily bypass isolated localized systems.**

***Index Terms*—Federated Learning, Distributed Energy Resources, Edge Computing, IDS , Cybersecurity.**

I. Introduction

Historically, electrical distribution networks operated as passive one-way systems that delivered power from substations to consumers. However, the rapid growth of Distributed Energy Resources (DERs) has changed this structure. The grid now behaves as a dynamic bidirectional cyber-physical system (CPS) [1]. Although this shift enables decentralized energy management, it also introduces instability. Weather conditions such as sunlight, wind, and temperature directly influence power generation, causing continuous and unpredictable changes in power flow. This creates a noisy and complex operating environment, making it harder to separate normal behavior from attacks and ultimately increasing the cyber-physical attack surface.

This inherent variability creates an ideal environment for sophisticated cyberattacks to remain undetected. Although conventional False Data Injection Attacks (FDIAs) often focus on manipulating sensor values directly, more advanced temporal attacks such as replay attacks exploit the timing and continuity of time-series data, making them significantly harder to detect within naturally volatile DER environments [9]. An attacker can record legitimate grid measurements during low-demand periods and replay them during peak demand to conceal malicious actions. Because the injected values remain within normal ranges, standard threshold-based detectors do not raise alarms. Instead, the attack takes advantage of the natural variability of DERs. From an isolated edge node's perspective, a replayed active power signal can be indistinguishable from natural fluctuations caused by weather variations such as cloud cover [10].

To secure decentralized grid infrastructure while preserving the privacy of sensitive operational data, researchers have increasingly adopted Federated Learning (FL) frameworks [11], [12]. However, existing FL-based intrusion detection methods remain largely topology-blind. They model DERs as independent entities and lack the spatial awareness needed to enforce physical constraints such as Kirchhoff's voltage and current laws across multi-node distribution networks [13]. As a result, temporally consistent but malicious signals can evade detection at the local level, even when they violate the physical consistency of the broader system.

To bridge the gap between IT-focused anomaly detection and OT-level physical constraint enforcement, this paper proposes the Intelligent Network Topology-Aware Collaborative Training **(INTACT)** framework. We utilized the IEEE 33-Bus distribution system [14] to simulate a neighborhood-level grid topology, specifically modeling the integration of active solar DERs to capture the complex dynamics of bidirectional power flow. The grid is partitioned into operational zones based on electrical proximity, where each zone is monitored by a localized edge computing client that trains a Long Short-Term Memory (LSTM) Autoencoder on its local physical state variables: Active Power (P), Reactive Power (Q), Voltage Magnitude ($|V|$) and Phase Angle (θ). Rather than centralizing raw grid data, zonal clients periodically transmit model weights to a federated aggregation server. Unlike standard Federated Averaging (FedAvg) [15], which treats all clients equally, our server assigns higher aggregation weight to zones that are more strongly connected in the physical network, measured by the admittance of the lines between them. This ensures that electrically coupled zones have greater influence on each other's learned model, reflecting the real physical relationships of the grid. Additionally, after each round of aggregation, the server compares reconstruction errors between neighboring zones. Since real physical disturbances

TABLE I: Comparison of Related Intrusion Detection Works

Method	Physical System Model	Attack Type	Key Limitation	Ref
Hybrid Federated Learning	None (communication layer only)	DoS & Data Integrity	Secures DNP3/Modbus protocols; no physical power flow modelling	[2]
Federated Deep Learning	Physical measurements (CPPS)	FDIAs	Does not quantify detection impact of DER-induced bidirectional flow	[3]
Vertical Federated Learning	Various sub-network measurements	FDIAs	No comparison of pre-DER vs. post-DER grid dynamics; unidirectional focus	[4]
Physics-Informed LSTM	FOC-VSM swing equations	FDIAs on Control Layer	Limited to grid-forming VSM control; not applicable to radial DER distribution networks	[5]
Physics-Informed Conv. AE	DER-integrated grids (P, Q, V, θ)	Stealthy FDIAs	Centralized data dependency; incompatible with federated edge deployment	[6]
Adversarial Autoencoder	DER-integrated unbalanced grids	FDIAs	Centralized architecture; raw data pooling incompatible with edge privacy	[7]
Federated LSTM Autoencoder	Substation sensor & IoT data	Generic anomalies	No power flow variables; topology-blind; no temporal attack evaluation	[8]
INTACT	**IEEE 33-Bus radial** (P, Q, $\|V\|$, θ) **with solar DERs**	**Temporal (Replay)**	**Combines federated privacy, topology-aware aggregation, DER bidirectional flow modeling, and temporal attack resilience.**	**Proposed**

affect multiple connected zones simultaneously, a Replay attack, which corrupts one zone while leaving its neighbors unaffected, is exposed by this spatial mismatch, detecting threats that local models alone cannot identify.

The specific contributions of this paper are as follows;

- **Multi-State Cyber-Physical Benchmarking:** The simulation and comparative analysis of a passive unidirectional baseline grid against an active bidirectional DER-integrated network, quantifying how dynamic power flow degrades the effectiveness of localized temporal anomaly detection.
- **Temporal Attack Modeling:** The synthetic injection of stealthy Replay attacks into a multi-variable IEEE 33-Bus dataset with active solar DERs, demonstrating that topology-blind models fail to reliably detect temporally consistent replay signals under natural DER variability.
- **Topology-Aware Federated Architecture:** A federated aggregation protocol in which client model contributions are weighted by inter-zone electrical admittance, encoding physical coupling directly into the learning process, augmented by a cross-zone reconstruction residual consistency check that identifies stealthy replay attacks by their absence of physically expected spatial propagation signatures, without centralizing raw operational data.

The remainder of this paper is organized as follows. Section II reviews related work on intrusion detection in cyber-physical power systems. Section III describes the cyber-physical simulation, dataset construction, and replay attack injection methodology. Section IV presents the proposed INTACT architecture and its topology-aware federated aggregation protocol. Section V evaluates the experimental results, and Section VI concludes the paper.

II. Related Work

Research on Intrusion Detection Systems (IDS) in cyber-physical power systems has generally developed in two distinct directions: federated learning approaches that prioritize communication privacy and physics-informed methods that leverage power flow constraints for anomaly detection. This section reviews the most relevant work along both areas and identifies the gap that motivates the proposed INTACT framework. Table I summarizes these studies, along with their key methods and limitations.

A. Federated Learning for Grid Intrusion Detection

Most FL approaches established the viability of distributed anomaly detection without centralizing the raw operational data. Purohit et al. [2] proposed FL-ADS, a federated anomaly detection system for DER networks targeting communication protocols such as DNP3 and Modbus. While the system effectively coordinates distributed clients, it secures the communication layer without modeling physical power flow variables, remaining blind to the physical coupling structure of the underlying network. Kausar et al. [3] developed a federated deep learning model using physical measurements for FDIA detection in cyber-physical power systems, yet do not isolate the vulnerability shift introduced when dynamic solar DERs are added to radial grids, leaving the impact of bidirectional power flow on detection performance unquantified. Kesici et al. [4] addressed data partitioning through vertical federated learning on the IEEE 123-bus and IEEE 37-node systems, achieving strong FDIA detection using various sub-network measurements. While the spatial-temporal split learning architecture represents a step toward physical awareness, the work focuses on static network partitioning and does not examine how the transition from passive unidirectional grids to active DER-integrated networks reshapes the detection landscape.

B. Physics-Informed Anomaly Detection

Physics-informed approaches have improved intrusion detection by integrating power system constraints directly into learning models. Khaleghi et al. [5] combined control-layer

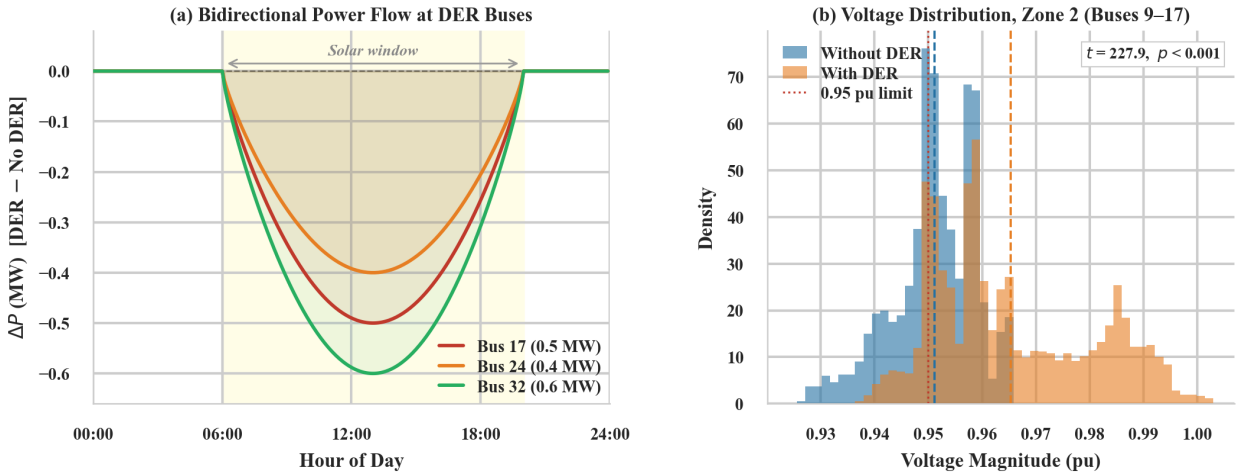

Fig. 1: Impact of solar DER integration on the IEEE 33-Bus system. (a) Mean diurnal active power difference at DER buses. (b) Voltage magnitude distribution in Zone 2 with and without DER ($t = 227.9$, $p < 0.001$).

grid physics with an LSTM model to improve cyberattack detection in virtual synchronous machines. While effective, this approach is limited to grid-forming control systems and does not generalize well to radial DER distribution networks. Jabbari Zideh and Solanki [6] further improved physics-aware intrusion detection using a multivariate convolutional autoencoder tested on DER-integrated IEEE 13-bus and 123-bus systems. Although effective against stealthy FDIAs, the model relies on centralized data collection, limiting its suitability for privacy-preserving federated edge environments.

C. Unsupervised Methods

Unsupervised learning has also been applied to DER-integrated power systems. Jabbari Zideh et al. [7] developed an adversarial autoencoder that effectively models DER-driven grid variability in unbalanced distribution networks. However, its centralized architecture requires raw data sharing, creating privacy and communication challenges that federated systems aim to eliminate. Most closely related to the architecture proposed here, Shrestha et al. [8] proposed a federated LSTM autoencoder for smart grid anomaly detection using substation sensor and IoT data at edge nodes. However, their model lacks physical power flow awareness, does not incorporate grid topology, and has not been evaluated against advanced temporal attacks such as replay.

D. Research Gap

Overall, existing research highlights that FL methods preserve privacy but lack physical topology awareness, while physics-informed approaches capture grid behavior but depend on centralized data. No current framework simultaneously combines federated privacy, topology awareness, DER bidirectional dynamics, and temporal attack detection within radial distribution networks.

III. Grid Simulation and Threat Modeling

A. IEEE 33-Bus Simulation Setup

The IEEE 33-Bus radial distribution system [16] is used as the simulation testbed. This benchmark network consists of a single feeder originating from a main feeder/slack bus (Bus 0, fixed at 1.0 pu) that supplies 32 load buses through a radial topology, making it representative of a neighborhood-level distribution network. All simulations are conducted using PandaPower [17], an open source power flow simulator.

To simulate DER integration, three solar photovoltaic units were placed at Bus 17 (0.5 MW), Bus 24 (0.4 MW), and Bus 32 (0.6 MW) across different network branches. Solar generation follows a normal weather-driven profile at 5-minute intervals and operates only between 06:00 and 20:00. The simulation covers four weeks, generating 8,064 timesteps per bus. At each timestep, four key physical variables are recorded: Active Power (P), Reactive Power (Q), Voltage Magnitude ($|V|$), and Phase Angle (θ). Two simulation scenarios are generated under identical load profiles and weather conditions:

- **No-DER baseline:** DER output is set to zero at all three injection buses, representing a passive unidirectional distribution network.
- **DER-active scenario:** All three solar units operate normally, introducing bidirectional power flow during solar generation hours.

The grid is divided into four operational zones based on electrical proximity and feeder structure, as defined in Table II.

TABLE II: Zone Partitioning of the IEEE 33-Bus System

Zone	Buses	DER Present
Zone 1	1–8	None
Zone 2	9–17	DER_1 at Bus 17 (0.5 MW)
Zone 3	18–24	DER_2 at Bus 24 (0.4 MW)
Zone 4	25–32	DER_3 at Bus 32 (0.6 MW)

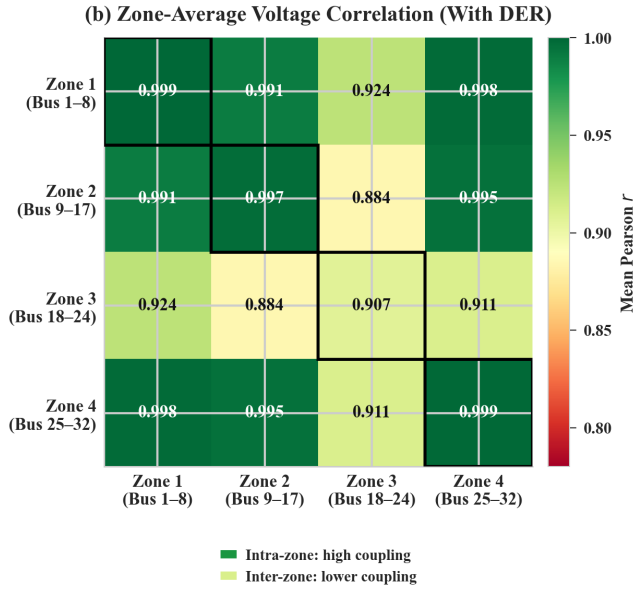

Fig. 2: Zone-average Pearson correlation of voltage magnitude under DER-active conditions. Bus 0 (slack bus) excluded.

B. Grid Comparison With & Without DER

To measure the impact of DER integration on the operating environment of anomaly detection systems, the two simulation scenarios are compared across voltage magnitude and active power dimensions. Fig. 1(a) shows the average daily active power difference (ΔP) at each DER bus, computed as the mean difference between the DER-active and No-DER scenarios across all 28 simulated days. During the solar window, all three DER buses exhibit strongly negative ΔP, with peak injections of -0.6 MW (Bus 32), -0.5 MW (Bus 17), and -0.4 MW (Bus 24) at approximately 12:00. Outside the solar window, $\Delta P = 0$ across all buses, confirming that the two scenarios are physically identical at night.

The effect on the voltage distribution is shown in Fig. 1(b). The No-DER distribution is narrow and concentrated near the lower operational limit (0.95 pu), reflecting the predictable voltage droop of a passive radial feeder. The DER-active distribution is significantly wider and shifted toward higher voltages, spanning from 0.93 pu to 1.00 pu. A Welch's t-test [18] confirms this difference is statistically significant ($t = 227.9$, $p < 0.001$), with DER integration raising the mean system voltage by $+0.008$ pu and reducing voltage violations below 0.95 pu by 68.8% (from 44,552 to 13,920 timesteps across the full system). This increased distributional spread elevates the baseline reconstruction noise floor for locally trained anomaly detectors, creating the operating environment in which temporally consistent replay attacks can exploit natural variability to evade detection.

C. Inter-Zone Coupling & Topological Heterogeneity

To validate the physical motivation for topology-aware federated aggregation, Fig. 2 presents the zone-average Pearson correlation of voltage magnitude time series across all four zones under DER-active conditions. Intra-zone correlations (diagonal) consistently exceed $r = 0.99$, indicating that buses within the same zone respond near-identically to network disturbances. Inter-zone correlations (off-diagonal) are lower, particularly involving Zone 3 ($r = 0.884$ with Zone 2, $r = 0.911$ with Zone 4), whose lateral branch topology produces voltage dynamics partially decoupled from the main feeder. This spatial heterogeneity in coupling strength motivates the use of inter-zone electrical admittance as aggregation weights in the proposed INTACT framework: zones that are more strongly coupled electrically should exert proportionally greater influence on each other's learned representations, a relationship that standard FedAvg ignores by treating all clients as equally weighted peers.

D. Stealthy Replay Attack Design

In a naive replay attack, an attacker substitutes real-time measurements with historical data via a hard cut. However, this introduces sudden temporal changes that localized time-series anomaly detectors, such as LSTM autoencoders, can easily identify. To rigorously evaluate the INTACT framework, we design a stealthy context-aware replay attack that remains consistent with local grid operating conditions while exploiting the spatial blind spot of zone-isolated detection.

Let the cyber-physical state of a zone at time t be the complete multi-variable measurement vector:

$$y(t) = [P(t),\ Q(t),\ |V(t)|,\ \theta(t)]^T \tag{1}$$

All four physical state variables — active power P, reactive power Q, voltage magnitude $|V|$, and phase angle θ — are simultaneously replaced at every timestep of the attack window. Full multi-variable substitution preserves the internal AC power flow consistency of the attacked zone: because all four variables are jointly replaced with historically coherent values, the local LSTM reconstructs the injected sequence with low residual $e(t) = \|y(t) - \hat{y}(t)\|^2 \leq \tau_{th}$, preventing detection by the isolated edge node.

The attack is executed in two steps:

- **Statistical Window Selection:** Rather than replaying arbitrary historical data, the attacker identifies a past time window whose statistical properties — mean and standard deviation across all features — closely match the target window [19]. For DER-equipped zones (Zones 2–4), targets are peak solar generation periods (10:00–14:00); for Zone 1 (load-only), targets are evening peak load periods (18:00–20:00). Source windows are constrained to differ by more than two hours from the target hour and to originate at least three days prior, preventing trivial same-hour matching in the synthetic dataset.
- **Cross-Fade Boundary Blending:** To mask the injection boundary, sensor readings are cross-faded over 12 timesteps (60 minutes) at each injection edge, smoothly transitioning from real-time to historical values and back [20]. Only the unblended middle portion is

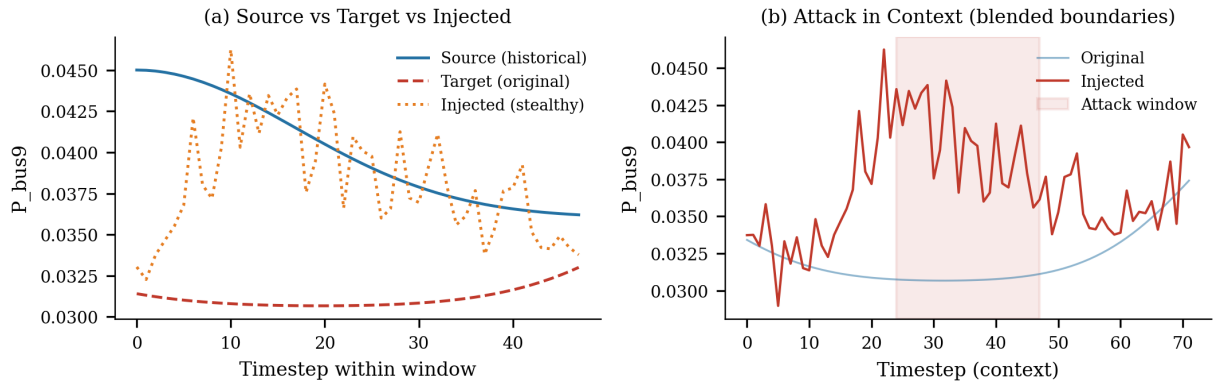

Fig. 3: Design of the stealthy replay attack for Zone 2. (a) Statistical window selection matching historical data to target operational ranges. (b) Cross-fade boundary blending to eliminate detectable temporal discontinuities.

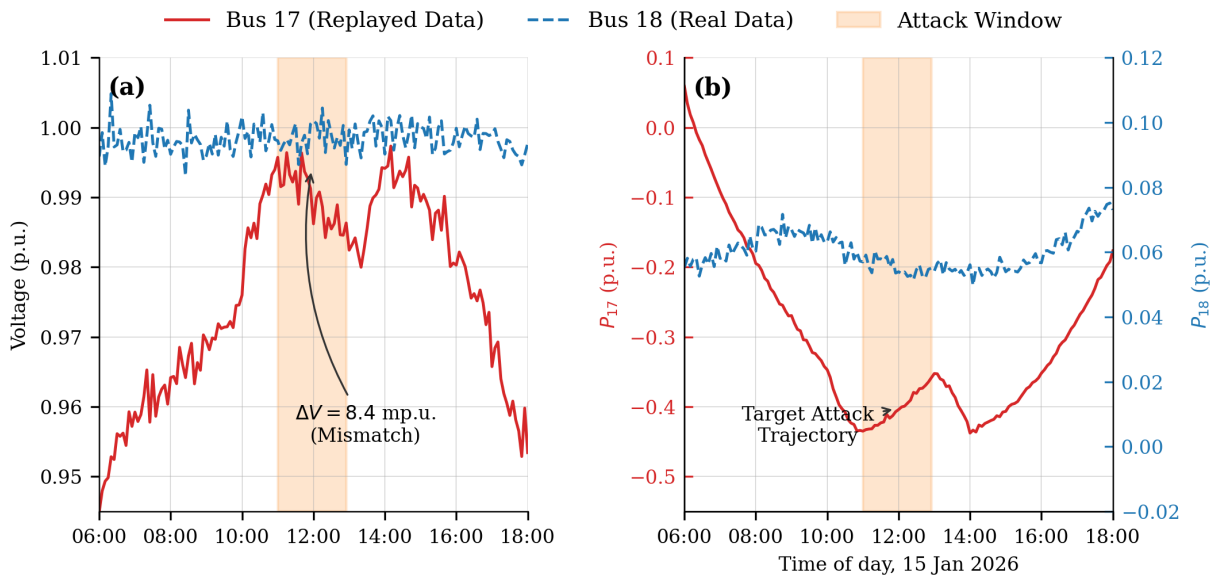

Fig. 4: Cross-zone spatial mismatch during a Zone 2 replay attack. The compromised node (Bus 17) reports replayed DER output that is physically inconsistent with the concurrent telemetry of its unaffected neighbor (Bus 18).

labeled as an attack event. This eliminates the sharp temporal discontinuity to which LSTM-based detectors are most sensitive [21]. Fig. 3 illustrates the result: Panel (a) shows that the source window shares a statistically similar operational range to the target, and Panel (b) shows the blended injection embedded in surrounding normal data with no detectable boundary.

- **Cross-Zone Spatial Inconsistency:** While the attack remains indistinguishable from a natural weather transient when examined locally, full multi-variable replacement in one zone necessarily breaks the spatial correlation between physically adjacent buses [9]. Fig. 4 illustrates this for a Zone 2 attack at 11:00. During the attack window, Zone 2 (Bus 17, compromised) reports replayed SCADA measurements consistent with a different historical operating period: voltage rises by $\sim$22 mp.u. above its pre-attack mean and active power injection drops to -0.40 p.u., characteristic of a high-DER-output interval several hours earlier. Meanwhile, Zone 3 (Bus 18, directly coupled to Bus 17 through feeder admittance) continues tracking the true real-time operating state — its voltage remains stable at $\sim$0.998 p.u. and its load remains unchanged at $\sim$0.055 p.u. If the DER at Bus 17 had genuinely shifted output by that magnitude, the admittance coupling between the two buses would have produced a measurable voltage and power change at Bus 18 — yet no such change occurs, exposing the data as replayed. This cross-zone divergence is invisible to the Zone 2 local model, which has no access to Zone 3 measurements, but constitutes a detectable admittance mismatch for any model observing both zones simultaneously. The INTACT framework

exploits this property through topology-aware federated aggregation, enabling the global model to learn expected inter-zone correlations during training and flag deviations at inference time even when local reconstruction residuals remain below the detection threshold.

E. Evaluation Dataset and Attack Injection Strategy

Stealthy replay attacks are injected independently into the test data of each zone: three events in Zone 1 and four events each in Zones 2–4, with each event spanning a 4-hour window (48 timesteps at 5-minute resolution). Each local model is evaluated exclusively on its own zone's attacked test file, which contains only the 32–36 bus-level features belonging to that zone.

For centralized and FL model evaluation, a combined test dataset is constructed by merging all four zone test files column-wise (128 features total). The global attack label is set to 1 whenever any zone is under attack, yielding 631 attack windows out of 4,002 total (15.8%). This evaluation protocol tests whether centralized and FL models can exploit cross-zone spatial correlations that are invisible to any single local model, providing a direct measure of the benefit conferred by global situational awareness.

IV. Isolated and Centralized Detection Baselines

We evaluate two baseline architectures to benchmark our INTACT framework. The first is an isolated local LSTM Autoencoder. Here, each model trains only on data from its own zone. The second is a centralized LSTM Autoencoder. This represents the theoretical upper bound with full data access. For both setups, the input sequences span $T = 30$ consecutive timesteps (2.5 hours). They cover four physical state variables per bus: P, Q, $|V|$, and θ.

A. Decentralized Local Autoencoder Training

The local configuration trains one LSTM Autoencoder per zone. It uses only that zone's measurement data. Zones do not share any information during training or inference. This mimics a scenario where each edge node operates as a completely isolated anomaly detector.

We ensured a mathematically fair baseline by optimizing each model independently. We used the Optuna framework for this step [22]. This allowed every local model to find its optimal architecture, such as hidden units and learning rates, tailored to its specific DER variance. Next, we calculated independent anomaly detection thresholds (τ_{th}) for each zone. These were set at a high percentile of the local training reconstruction errors, ranging from the 99.1^{th} to the 99.9^{th} percentile based on local noise profiles.

During testing, any reconstruction error above the localized τ_{th} triggers an alarm. Table III shows how each zone performed against the stealthy replay attack.

TABLE III: Localized Baseline Detection Performance

Zone	Thresh.	Prec.	Recall	F1	AUC
Zone 1	0.0067	0.44	0.52	0.47	0.848
Zone 2	0.0045	0.44	0.69	0.54	0.903
Zone 3	0.0059	0.74	0.75	0.74	0.923
Zone 4	0.0033	0.68	0.94	0.79	0.985
Avg	–	**0.58**	**0.73**	**0.64**	**0.915**

Detection performance varies widely across the zones. This reflects the diverse operating conditions caused by DER integration. Zone 4 hosts the largest DER unit (0.6 MW at Bus 32). It achieves the highest recall at 0.94. This happens because suppressing a large solar signal creates a very obvious statistical drop. In contrast, Zone 2 contains a smaller DER (0.5 MW at Bus 17). It only achieves an F1-Score of 0.54. The natural voltage swings from the solar panel raise the reconstruction noise floor, hiding the attack. Zone 1 has no DER at all and performs the worst (F1-Score: 0.47). The attack here targets the evening load peak, which easily blends into normal demand fluctuations.

Fig. 5 visualizes the local LSTM performance for Zone 2. Panel (a) shows a massive overlap between normal and attack reconstruction errors. The attack windows barely push past the detection threshold. Panel (b) confirms this timeline. Three out of four attacks stay below or just barely cross the threshold line. We also see a false positive spike near window 720. This reflects a normal operational change such as natural variations in solar generation or load demand that the model incorrectly classified as an anomaly.. These results confirm a major flaw. Even when perfectly tuned, isolated local detectors cannot catch stealthy replays hidden inside natural DER variability.

B. Centralized ML Model Training

The centralized baseline takes a different approach. It trains a single LSTM Autoencoder on the combined raw data from all four zones at once. The input space spans 128 features at each timestep. This gives the model complete visibility over the entire physical system.

This model processes a much larger feature space. Therefore, we ran a dedicated Optuna search on the combined dataset to find the optimal global architecture. The best setup used two stacked LSTM layers (32 and 64 units), no dropout, and a learning rate of 5.73×10^{-3}. We set the anomaly threshold at the 99.4^{th} percentile ($\tau_{th} = 0.0046$) of the combined training errors. We must ensure a fair comparison with the federated models later. Therefore, we locked this exact global architecture and threshold percentile for all subsequent Federated Learning (FL) experiments.

This centralized setup represents the theoretical performance ceiling. It assumes unrestricted server-side access to all operational measurements while disregarding the stringent cybersecurity, access control, and information-protection requirements imposed in real-world grid environments. In practice, such architectures face significant deployment barriers due to regulatory standards such as NERC CIP-005 and CIP-011, which enforce strict Electronic Security Perimeter controls and protection of Bulk Electric System (BES) Cyber System Information, thereby

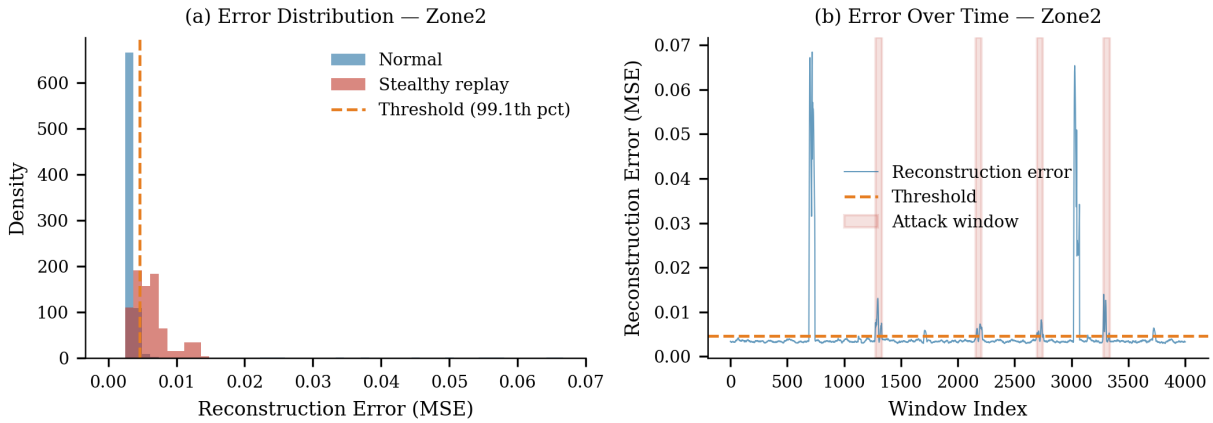

Fig. 5: Localized LSTM Autoencoder performance on Zone 2.

limiting unrestricted sharing of raw operational telemetry across organizational boundaries [23], [24].

The centralized model achieved an average F1-Score of 0.77 and an AUC of 0.928. This easily beat the local average F1-Score of 0.64. This 13-point jump proves a critical concept. Cross-zone information is vital for catching stealthy attacks. When the model sees the entire system at once, it can flag localized anomalies that look perfectly normal in isolation.

However, the centralized model has a drawback. It incurs a higher false positive rate (FPR = 0.060) compared to the local models (FPR = 0.030). The massive 128-feature input captures too much inter-zone noise. This occasionally triggers false alarms during normal, grid-wide weather events. This exact tradeoff motivates our INTACT framework. We want to replicate the high detection rates of central awareness, but keep the per-zone data locality to prevent unnecessary false alarms.

V. Proposed Topology-Aware Federated Learning

A. Limitations of Standard Federated Aggregation

B. Admittance-Weighted Aggregation (INTACT)

C. Cross-Zone Consistency Verification

VI. Experimental Results and Discussion

A. Global Architecture Comparison

B. Analysis of Spatial Mismatch Detection

VII. Conclusion

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TABLE IV: Global Architecture Performance Comparison

Architecture	Privacy Preserved	Topology Aware	Avg Precision	Avg Recall	Avg F1-Score	AUC
Local LSTMs	Yes	No	0.58	0.73	0.64	0.915
Centralized ML	No	Yes	0.72	0.83	0.77	0.928
FedAvg (20R/3E)	Yes	No	0.64	0.21	0.32	0.831
FedProx ($\mu = 0.001$)	Yes	No	0.66	0.20	0.31	0.815
FedProx ($\mu = 0.01$)	Yes	No	0.63	0.19	0.29	0.816
FedAdam ($\eta = 0.01$)	Yes	No	0.61	0.22	0.33	0.811
INTACT (Proposed)	**Yes**	**Yes**	**[Insert]**	**[Insert]**	**[Insert]**	**[Insert]**

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